

Campus PollLive

Fall 2025

Agneya Limaye

1. Project description

The purpose of this project is to develop a fully functional web-based polling application that enables registered users to create polls, vote on existing polls, and view real-time results. The application provides an interactive platform where users can share their opinions on topics such as campus life, courses, student interests, and general discussions. By focusing on clarity, responsiveness, and ease of use, the system demonstrates the complete lifecycle of a dynamic web application using the core concepts taught in COMP 3540.

The system incorporates secure user authentication through Sign Up, Sign In, and Sign Out, with PHP sessions ensuring authenticated access. Beyond login functionality, the application implements more than eight user-level features, including updating profile information, deleting an account, creating polls, voting, browsing polls, searching and filtering, viewing personalized “My Polls” results, and automatically updating poll results in real time. All dynamic interactions and updates occur without refreshing the entire page, achieving a smooth Single Page Application (SPA) experience.

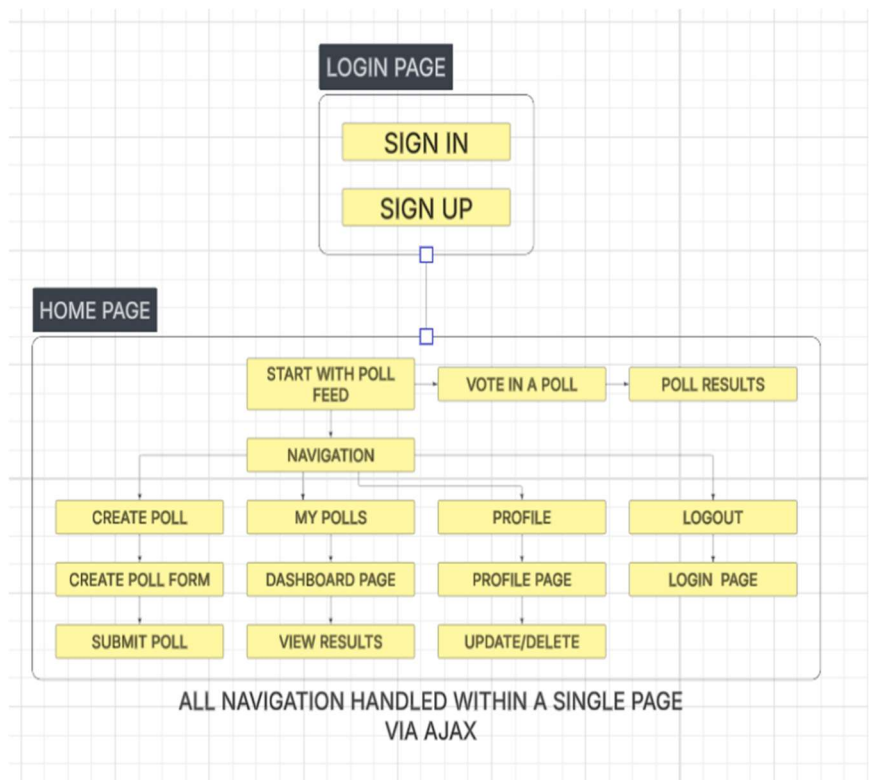
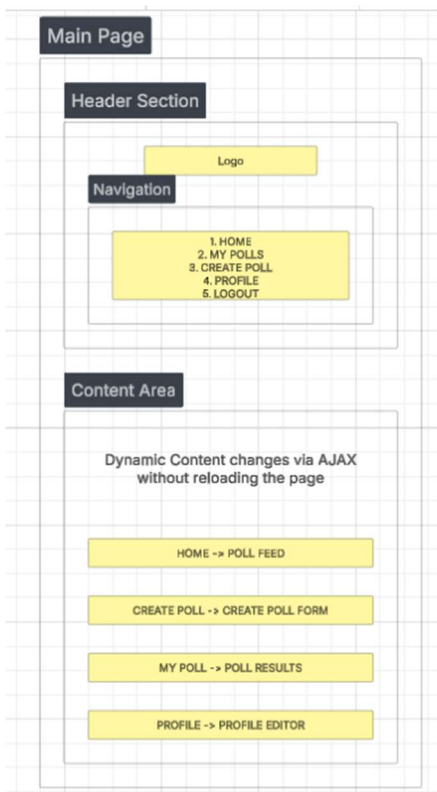
The application follows an MVC-style architecture, separating the data model, controller logic, and view templates for maintainability. AJAX requests (using plain JavaScript XMLHttpRequest) handle all dynamic updates such as searching, voting, and automatic 5-second refreshes. Bootstrap is used for layout and styling, ensuring a clean and responsive interface. A MySQL database provides persistent storage for users, polls, options, and votes, enabling the entire system to be tested online as required by the course.

2. List of detail user level, not programming level, features with descriptions

User-level features	Descriptions
Update User Profile	Allow a logged-in user to change their display name, email address, and short bio. The values are also updated in the database using AJAX. Display Name and Bio, right now are reflected only on the profile.
Delete User Account	Let users permanently delete their account. All polls created by the deleted user remain visible but are marked “by deleted user.”
Creating a New Poll	Enables users to create a poll by entering a question, along with 4 answer options, and a category – “Campus”, “Courses”, or “Sports”. Polls are saved to the database and appear instantly in the home feed.

Browse active Polls	Displays all currently active polls in a live feed ordered by creation time. The list dynamically refreshes every 5 seconds to display any new polls.
Vote in Polls	Allow each logged-in user to select one option in any active poll. The system checks that the user hasn't already voted in that poll. Votes are recorded via AJAX and update the results instantly.
View Poll Results	After voting, users can view a progress-bar-style result chart that updates in real time when other users vote. Percentages and totals are computed from the database using AJAX.
View "My Polls" Dashboard	Provides a personalized dashboard where users can see and manage all polls they created, including current results.
Search and Filter Polls	Let users search for polls by keywords or filter them by category. Results load dynamically without reloading the page.

3. Page layout diagrams, and user interface



4. Test procedures

Note. Update the table whenever you complete/update a user-level feature.

User-level features	Descriptions	Expected Result
Sign up	Click signup button -> enter form -> click submit->signin	User is redirected to the Sign In page (or logged in) and can later sign in using the new credentials. If username already exists, an error message is shown.
Sign in	Enter valid username and password -> click submit -> main page	User is redirected to the Main Page, header shows "Logged in as <username>", and navigation buttons (Home, My Polls, Profile, Sign Out) become available.
Sign Out	Click sign out nav button -> click submit -> start page	Session ends, user is taken back to the Start Page. Reloading the Main Page URL no longer shows the poll interface without logging in again.
Update User Profile	Click Profile -> change email, display name, and bio -> click Save Profile.	A confirmation message appears. Returning to the Profile page (or reloading Main Page) shows the updated email/display name/bio. Display name is only recorded for profile as of now.
Delete User Account	Profile -> click Delete Account, confirm when prompted -> redirected to start page-> try signing in again	User is logged out and can no longer sign in with that account. Existing polls remain in the system.
Creating a New Poll	On Home -> Click "Create A Poll" -> Enter question, options, and category -> Submit -> Check that the new poll appears immediately in "My Polls" and the public feed.	A success message appears on the create a poll form, and the new poll shows up at the top of the "All Polls" feed and in My Polls. The poll is also visible from another account.

View My Polls	While logged in, click My Polls. Make sure you have created at least one poll.	Only polls created by the logged-in user are shown. Each poll displays question, description, category, and current results (including progress bars) and reflects any new votes.
Browse active Polls	Log in -> Go to "Home" -> Confirm a list of active polls is displayed and updates dynamically when a new poll is created in another session.	
Vote in Polls	Select any active poll -> Choose one option and click "Vote" -> see the notification of vote recorded	The page does not fully reload; instead, the poll card switches to a results view via AJAX. Votes and percentages update.
View Poll Results	After voting scroll down to view the recorded result along with the overall results with other votes and percentages, and a progress bar representing the percentage	Each option is displayed with total votes, percentage, and a Bootstrap progress bar whose length matches the percentage. Progress bars update after new votes.
View "My Polls" Dashboard	Open "My Polls" -> Confirm all polls created by the current user are listed with accurate vote counts and correct status (active/closed).	
Search and Filter Polls	On home, create 2 forms with different keywords and categories -> use search to filter keywords and category bar to filter categories (Campus, Courses, Sports)	The poll list updates without reloading the whole page, showing only polls that match the keyword/category. Clicking Clear resets, the list to all polls.